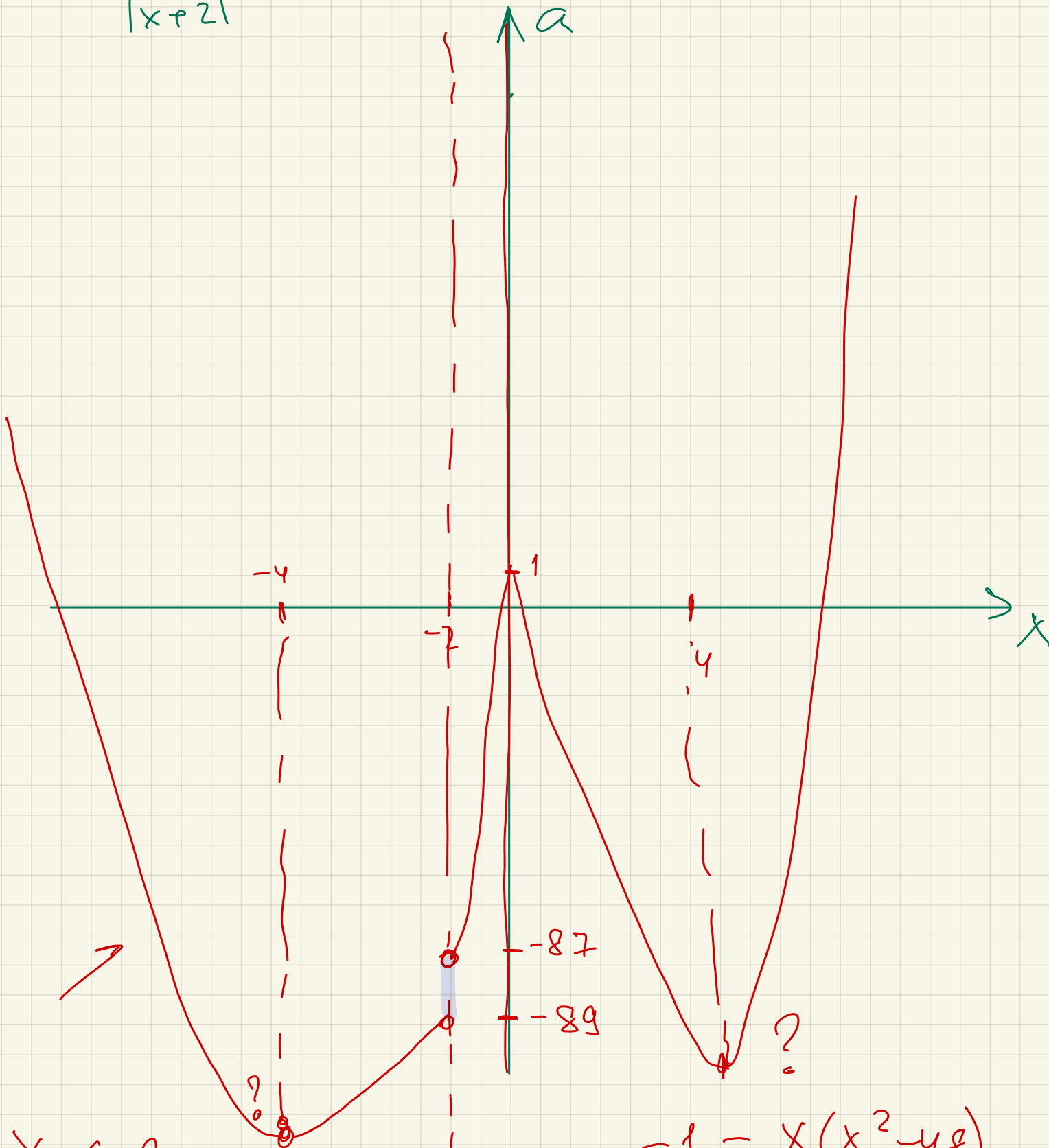




$$\frac{x+2}{|x+2|}$$

$$+ |x|(x^2 - 48) = a$$

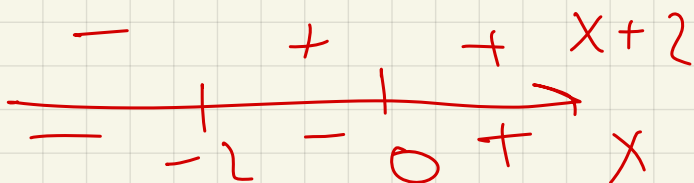
$\uparrow a$



$$x < -2$$

$$-1 - x(x^2 - 48)$$

$$-x^3 + 48x - 1 = a$$

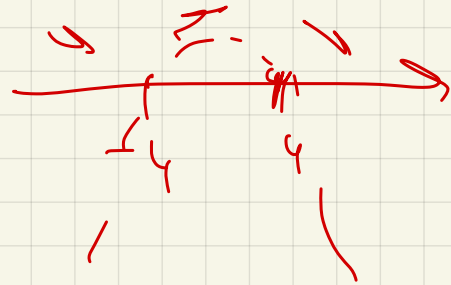


$$a(-2) = +8 - 48 \cdot 2 - 1 =$$

$$a' = -3x^2 + 48$$

$$x^2 = \frac{48}{3} = 16$$

$$x = \pm 4$$

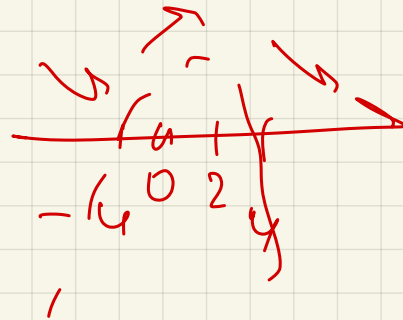


$$x \leq 0$$

$$1 - x(x^2 - 48) = a$$

$$a(-2) = +8 - 96 + 1 = -87$$

$$a' = -3x^2 + 48$$



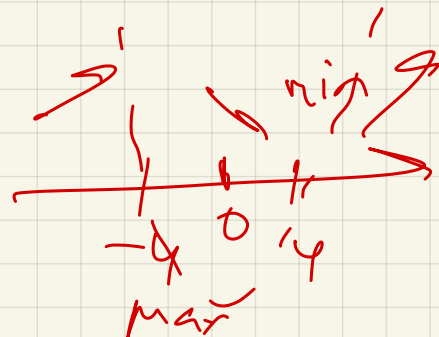
$$a(6) = 1$$

$$x > 0$$

$$1 + x(x^2 - 48) = a$$

$$x^3 - 48x + 1 = a$$

$$a' = \underline{3x^2 - 48}$$



$$y = (2a-3)x + (x-a)|x+3|$$

$$x \geq -3$$

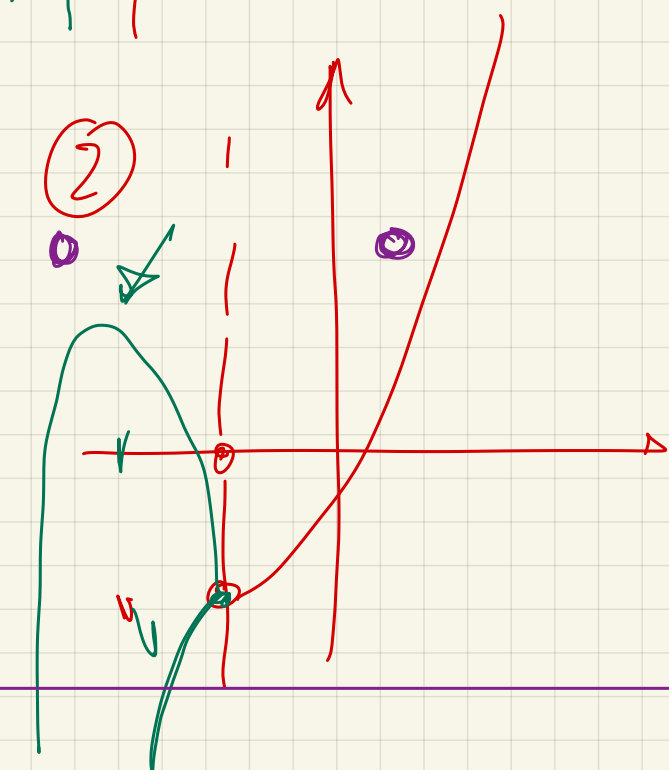
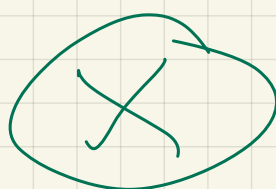
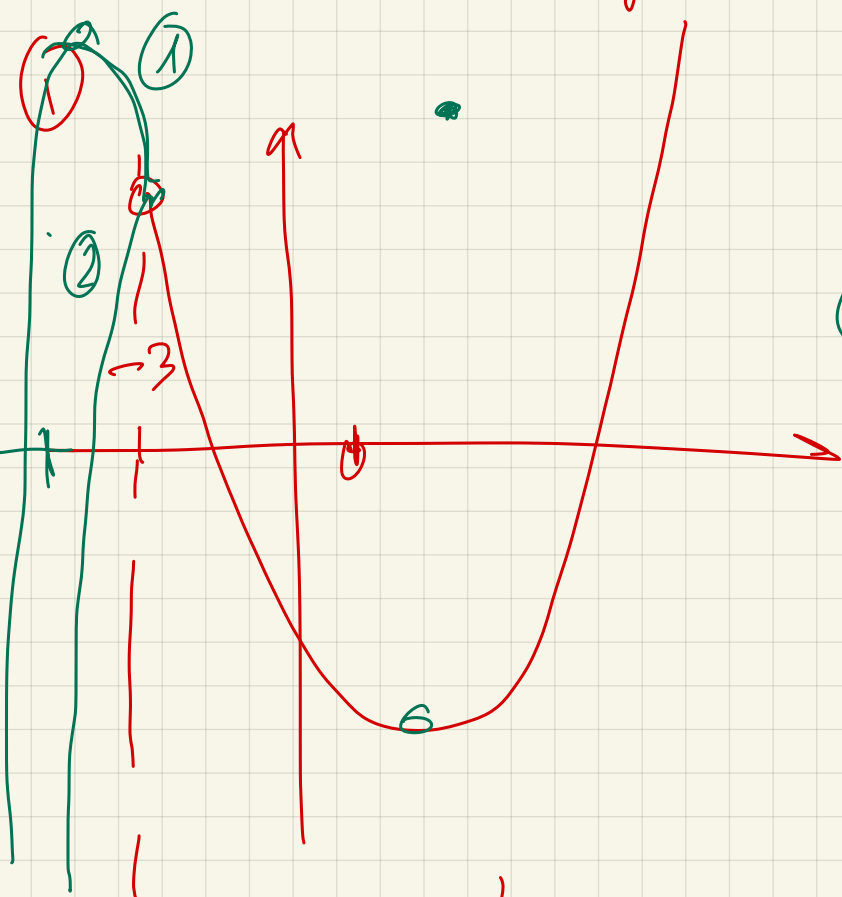
$$y = x^2 + ax - 3$$

$$x_0 = -\frac{a}{2}$$

$$x < -3$$

$$y = -x^2 + (3a-6)x + 3a$$

$$x_0 = \frac{3a-6}{2}$$

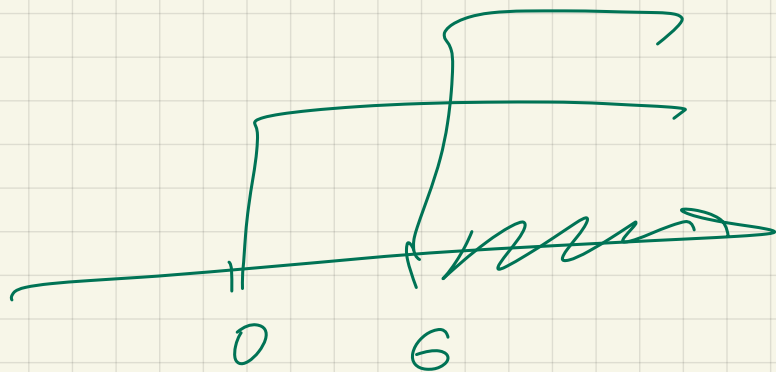


$$\begin{cases} x_0 \leq -3 \\ x \geq -3 \end{cases}$$

$$-\frac{a}{2} \leq -3$$

$$\frac{a}{2} \geq 3$$

$$\frac{a}{4} \geq 6$$



$$\frac{3a-6}{2} \geq -3$$

$$3a-6 \geq -6$$

$$3a \geq 0$$

$$a \geq 0$$

$$a \in [6; +\infty)$$

$$y = \frac{a}{2}x + a$$

$$12 \geq 6 > 6$$

$$y = a|x| - \left|\frac{a}{2}\right|$$

$$1) a = 0$$

$$y = 0$$

$$y = 0$$

$\emptyset$

$$2) a < 0$$

$$y = \frac{a}{2}x + a$$

$$a = -2$$

$$ax + \frac{ax}{2} = \frac{a}{2} - a$$

$$\frac{3ax}{2} = -\frac{a}{2} \quad x = -\frac{1}{3}$$

$$-ax + \frac{a}{2} = \frac{a}{2}x + a$$

$$\frac{3ax}{2} = -\frac{a}{2}$$

$$ax + \frac{a}{2} = \frac{a}{2}x + a$$

$$y = a|x| + \frac{a}{2}$$

$$y = -x - 2$$

$$y = -2|x| - 1$$

$$\frac{a}{2}x = \frac{a}{2}$$

$$x = 1$$

$$ax + \frac{a}{2} = \frac{a}{2}x + a$$

$$ax - \frac{a}{2} = a - \frac{a}{2}$$

$$\frac{ax}{2} = \frac{a}{2}$$

$$x = 1$$

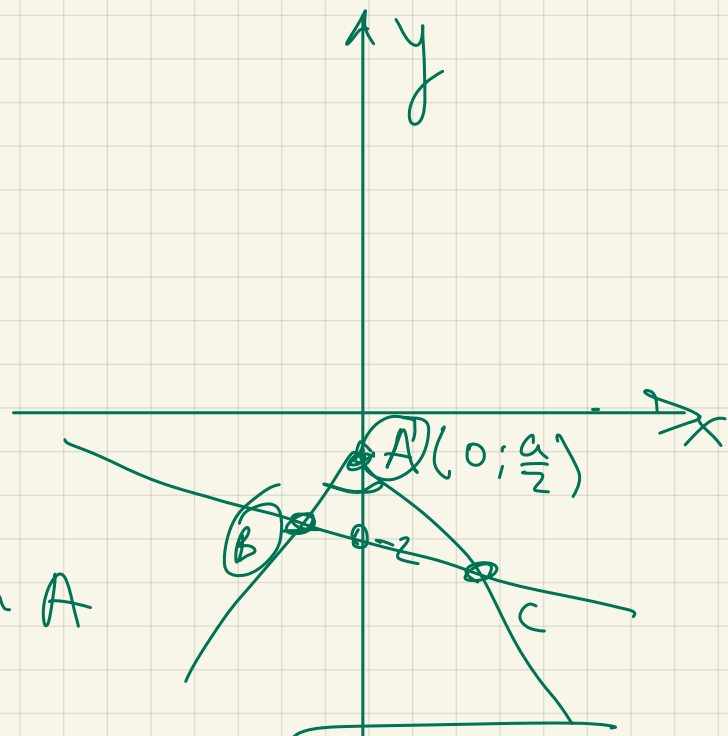
$$B \left(-\frac{1}{3}; \frac{5a}{6} \right) ?$$

$$C \left(1; \frac{3a}{2} \right) ?$$

$$A \left(0; \frac{a}{2} \right)$$

$$\begin{cases} AB = \\ AC = \end{cases}$$

$$S' = \frac{1}{2} \underline{AB \cdot AC} \cdot \sin A$$



$$S = \frac{1}{2} \frac{\sqrt{1+c^2} \sqrt{1+c^2} \left| \frac{-2c}{1+c^2} \right|}{3} =$$

$$12 \geq \left(-\frac{a}{3} \right) > 6$$

$$-\frac{a}{3} \leq 12$$

$$\frac{a}{3} \geq -12$$

$$a \geq -36$$

$$-\frac{a}{2} > 6$$

$$\frac{a}{2} < -6$$

$$a < -12$$

$$A(x_0; y_0)$$

$$B(x_1; y_1)$$

$$|AB| = \sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2}$$

$$AC = \sqrt{1^2 + \left(\frac{a}{2} - \frac{3a}{2} \right)^2} = \sqrt{1 + a^2}$$

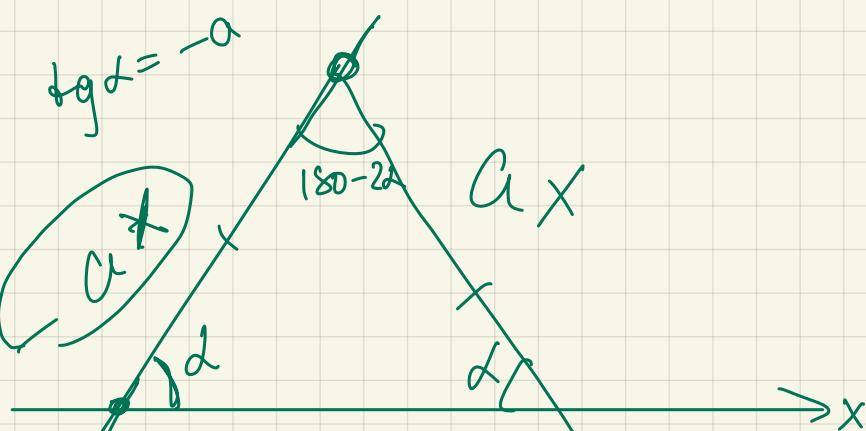
$$AB = \sqrt{\left(0 + \frac{1}{3} \right)^2 + \left(\frac{a}{2} - \frac{5a}{6} \right)^2}$$

$$= \sqrt{\frac{1}{9} + \left(\frac{2a}{6} \right)^2} =$$

$$\sqrt{\frac{1}{9} + \frac{4a^2}{36}} =$$

$$\sqrt{\frac{4 + 4a^2}{36}} =$$

$$\frac{2\sqrt{1+a^2}}{6} = \frac{\sqrt{1+a^2}}{3}$$



$$\sin(180 - 2\alpha) = \sin 180 \cos 2\alpha - \sin 2\alpha \cos 180$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\lg 2 = -a$$

$$\frac{\sin x}{\cos x} = -a \Rightarrow \sin x = -a \cos x$$

$$\sin 2\alpha = 2(-a \cos \alpha) \cos \alpha = -2a \cos^2 \alpha$$

$$1 + \lg^2 2 = \frac{1}{\cos^2 2}$$

$$\cos^2 \alpha = \frac{1}{1 + \tan^2 \alpha} = \frac{1}{1 + e^2}$$

$$\sin 2\alpha = \frac{2a}{1+a^2}$$

$$a > 0$$

$$y = a|x| - \frac{a}{2}$$

$$y = \frac{a}{2}x + a$$

$$A \left(0; -\frac{a}{2} \right)$$

$$B \left(-1; \frac{a}{2} \right)$$

$$C \left(3; \frac{5a}{2} \right)$$

$$\sin 2\alpha = \frac{2a}{1+a^2}$$

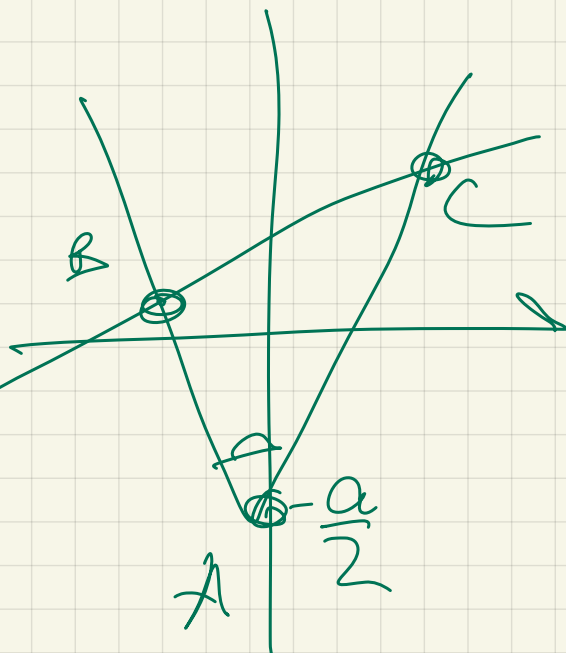
$$\sqrt{ax - \frac{a}{2}} = \frac{a}{2}x + a$$

$$-ax - \frac{a}{2} = \frac{a}{2}x + a$$

$$ax - \frac{ax}{2} = \frac{a}{2} + a$$

$$\frac{ax}{2} = \frac{3a}{2}$$

$$x = 3$$



$$AB = \sqrt{1^2 + a^2}$$

$$AC = \sqrt{4^2 + (2a)^2} = 2\sqrt{1+a^2}$$

$$x \rightarrow 3 \quad y \rightarrow$$

$$x \rightarrow -1 \quad y \rightarrow$$

$$\frac{ax}{2} + ax = -\frac{a}{2} - a$$

$$\frac{3ax}{2} = -\frac{3a}{2}$$

$$x = -1$$

$$6 < \frac{\sqrt{1^2 + a^2} + \sqrt{1 + a^2}}{2} \quad \frac{2a}{1+a^2} \leq 12$$

$$6 < a \leq 12$$